REVIVING CIRCULANT PRECONDITIONERS FOR ADAPTIVE MESH REFINEMENT *

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Abstract. We present a preconditioner for solving fractional differential equations (FDEs) on an adaptive mesh. Adaptive refinement of the problem domain results in a stiffness matrix with Toeplitz blocks along the main diagonal, while the FDE yields a dense stiffness matrix, where off-diagonal blocks are stored as low-rank approximations. Our adaptive preconditioner utilizes ideas from the circulant preconditioner of Chan and Strang [SIAM Journal on Scientific Computing, 1989], which takes advantage of the Toeplitz blocks on the diagonal and accounts for the low-rank nature of the off-diagonal blocks. We demonstrate its effectiveness at accelerating convergence for our systems and emphasize its efficient application. This work presents theoretical results about the spectral clustering of the preconditioned system. In order to prove these results, special consideration is taken on how the low-rank blocks perturb the eigenvalues of the Toeplitz block-diagonal system. Numerical tests are used to inspect any assumptions and validate our results.

Key words. Preconditioner, Adaptive Refinement, Finite-element Method, Toeplitz, Circulant, DST

1. Introduction.

In this paper we consider fractional differential equations (FDEs) of the form

$$D_x^\alpha u(x) = f(x),\ x \in (a,b), \tag{1.1}$$

where D_x^α is the Riesz fractional derivative of order $1 < \alpha < 2$ and $f \in L^2([a,b])$.

Many numerical solvers for FDEs discretize the domain on a uniform mesh; the uniform mesh results in a linear system with a Toeplitz matrix. Still FDEs present unique difficulties in numerical methods due to the non-locality of the operator resulting in a dense stiffness matrix. Additionally, as part of the fractional derivative the singular kernel, $k(x,s) = |x-s|^{1-\alpha}$, is integrated, resulting in end-point singularities which make Galerkin finite-element methods sub-optimal [9, 12]. To address this, Jin and Zhou developed a reconstruction technique that isolates the singularity to improve FEM convergence [7]. Alternatively, Wang and Du developed a Toeplitz preconditioner that reduces iteration cost to $\mathcal{O}(n \log n)$ using the preconditioned conjugate gradient squared method [12].

The above methods, however, require uniform meshes. An optimal mesh would have the L^∞ error equally distributed on each interval, but a uniform mesh does not in general satisfy this condition. Indeed, the singularities of FDEs force higher error in the intervals near the boundary. Alternatively, to achieve desired accuracy across the whole domain we use use an adaptive mesh and adaptive finite-element method (AFEM) [10, 13]. The usual AFEM algorithm iterates through the process

$$\texttt{SOLVE} \to \texttt{ESTIMATE} \to \texttt{MARK} \to \texttt{REFINE}, \tag{1.2}$$

but the computational cost is dominated by the `SOLVE` step [13].

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Our aim is to build a preconditioner for the stiffness matrix, A, from an AFEM discretization of Equation (1.1) that will reduce this work. We investigate what structure the adaptive mesh retains and how that is reflected in the structure of A. Then, using preconditioning techniques formulated for Toeplitz matrices [3, 4, 5, 6], we build a preconditioner, show the preconditioned system has eigenvalues clustered around 1 and demonstrate numerical results with superlinear convergence.

2. Background.

2.1. Properties of Structured Matrices. There has long been interest in solving Toeplitz linear systems efficiently. A matrix A is called Toeplitz if $a_{ij} = a_{i-j}$. Arbitrary $n \times n$ matrices have up to n^2 entries and, in general, arbitrary linear systems are solved directly in $\mathcal{O}(n^3)$ FLOPS. Since a Toeplitz matrix has at most $2n-1$ unique entries, we expect to solve Toeplitz systems with $\mathcal{O}(n^2)$ FLOPS, (see for example Levinson's algorithm, Trench's algorithm). Even with these improvements, however, fast Toeplitz algorithms are unstable in general and infeasible for large systems that arise in application [2]. Instead we turn to iterative Krylov and multigrid methods [6, 9, 13]. For these methods we take advantage of Toeplitz structure by using circulant preconditioners.

Toeplitz matrices commonly arise in partial differential equations discretization, FDEs, signal processing, and control theory. Often the Toeplitz matrices are also symmetric positive definite (SPD). Given an SPD Toeplitz system $Lx = b$, the idea introduced by Strang and Chan [4] is to use certain circulant preconditioners C so that $C^{-1}Tx = C^{-1}b$ is solved in fewer iterations. A circulant matrix is a Toeplitz matrix, that additionally has the "wrap-around" property where the last entry of each row is the first entry of the subsequent row:

$$L = \begin{pmatrix} \ell_0 & \ell_{-1} & \cdots & \ell_{-(n-1)} \\ \ell_1 & \ell_0 & \cdots & \ell_{-(n-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{n-1} & \ell_{n-2} & \cdots & \ell_0 \end{pmatrix} \quad C = \begin{pmatrix} c_0 & c_1 & \cdots & c_{n-1} \\ c_{n-1} & c_0 & \cdots & c_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ c_1 & c_2 & \cdots & c_0 \end{pmatrix} \tag{2.1}$$

Equation 1: L is a Toeplitz matrix and C is circulant.

As will be discussed in Section (2.2), the stiffness matrix resulting from AFEM applied to Equation (1.1) is not Toeplitz, but does contain SPD Toeplitz blocks. Toeplitz matrices have many special properties, but we focus on their connection to functions in the Wiener class.

Definition 2.1 ([4]). *Given a singly infinite, symmetric Toeplitz matrix L we define the finite subsection L_m to be the $m \times m$ principle submatrix. That is,*

$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \\ \ell_1 & \ell_0 & \ell_1 & \ddots \\ \ell_2 & \ell_1 & \ell_0 & \ddots \\ & \ddots & \ddots & \ddots \end{pmatrix} L_m = \begin{pmatrix} \ell_0 & \ell_1 & \cdots & \ell_{m-1} \\ \ell_1 & \ell_0 & \cdots & \ell_{m-2} \\ \vdots & \vdots & \ddots & \vdots \\ \ell_{m-1} & \ell_{m-2} & \cdots & \ell_0 \end{pmatrix}.$$

Assume $\sum_{k=-\infty}^{\infty} |\ell_k| < \infty$*, then the matrices induce functions* $\tilde{\ell}(z) = \sum_{k=-\infty}^{\infty} \ell_k z^k$ *and* $\tilde{\ell}_m(z) = \sum_{k=-(m-1)}^{m-1} \ell_k z^k$ *are real, positive, and in the Wiener class for* $|z| = 1$*.*

Previous work on Toeplitz systems offers a few options for circulant preconditioners. Although their construction differs, the spectrum of the preconditioned systems are asymptotically the same [3]. Following [1], given a symmetric Toeplitz matrix L we build a Hankel correction, H, where

(2.2)
$$L = \begin{pmatrix} \ell_0 & \ell_1 & \ell_2 & \cdots & \ell_{n-2} & \ell_{n-1} \\ \ell_1 & \ell_0 & \ell_1 & \cdots & \ell_{n-3} & \ell_{n-2} \\ \ell_2 & \ell_1 & \ell_0 & \cdots & \ell_{n-4} & \ell_{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \ell_{n-2} & \ell_{n-3} & \ell_{n-4} & \cdots & \ell_0 & \ell_1 \\ \ell_{n-1} & \ell_{n-2} & \ell_{n-3} & \cdots & \ell_1 & \ell_0 \end{pmatrix} H = \begin{pmatrix} \ell_2 & \ell_3 & \ell_4 & \cdots & \ell_{n-1} & 0 & 0 \\ \ell_3 & \ell_4 & \ell_5 & \cdots & 0 & 0 & 0 \\ \ell_4 & \ell_5 & \ell_6 & \cdots & 0 & 0 & \ell_{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \ell_5 & \ell_4 & \ell_3 \\ 0 & 0 & \ell_{n-1} & \cdots & \ell_4 & \ell_3 & \ell_2 \end{pmatrix}.$$

Then define $\boldsymbol{\tau} := L - H$. Both L and H are symmetric, so they are completely represented by at most n unique values: the diagonals of $L : \ell_0, \ell_1, \ldots, \ell_{n-1}$, and the anti-diagonals of $H : \ell_2, \ell_3, \ldots, \ell_{n-1}, 0, 0$.

The following lemma summarizes properties of $\boldsymbol{\tau}$.

LEMMA 2.2 (see [1]). *Let L be a Toeplitz matrix and H the corresponding Hankel correction as in Equation (2.2). Then $\boldsymbol{\tau} = L - H$ has the following properties:*

1. *$\boldsymbol{\tau}$ is diagonalized by the type-I discrete sine transform (DST) matrix, S. We write $\boldsymbol{\tau} = S\Lambda S^{-1} = S\Lambda S$.*
2. *$\boldsymbol{\tau}$ can be applied in $\mathcal{O}(n \log n)$ FLOPS.*
3. *Let L_1 be the first column of L. Then, the k-th eigenvalue of $\boldsymbol{\tau}$ is proportional to the k-th entry of SL_1. Specifically,*
$$\lambda_k(\boldsymbol{\tau}) = c_k [SL_1]_k, \text{ where } c_k := \sqrt{\frac{n+1}{2}} \frac{1}{\sin(\frac{\pi k}{n+1})}. \tag{2.3}$$
4. *Each eigenvalue of $\boldsymbol{\tau}$ is written as the truncated function $\tilde{\ell}_m$ evaluated somewhere on the unit circle, $\lambda_k(\boldsymbol{\tau}) = \tilde{\ell}_m(z_k)$, where $|z_k| = 1$.*

2.2. AFEM stiffness matrix. Again, let A be the stiffness matrix resulting from AFEM applied to Equation 1.1. During the `MARK` module of AFEM (1.2), it is often the case that adjacent elements have similar error estimates, and are either both marked or both not marked for refinement. As a result, after several refinements there are still subsections of the domain that have been refined to the same level. We refer to these subdomains as *locally uniform*. Since uniform meshes give rise to SPD Toeplitz systems, if we formed the stiffness matrix for a locally uniform subdomain we would have an SPD Toeplitz matrix. So wherever there are adjacent elements of the same size we find a corresponding Toeplitz block on the main diagonal of our stiffness matrix, A. The size of this block depends on how many adjacent elements are the same size. In practice we have observed that the boundary of the domain almost always requires the greatest level of refinement. In general we have larger Toeplitz blocks from locally uniform subdomains in the middle of the matrix, and smaller blocks—or indeed 1×1 blocks—near the boundary. We use A_D to denote the block diagonal with Toeplitz blocks part of A.

As mentioned previously, the fractional derivative results in a dense matrix, but we observe off-diagonal decay, as elements that are physically separated in the domain

have weaker interaction. The rate of decay depends on the fractional order α. Blocks that are most distant from the diagonal contain less information and we assume they have small-norm. We denote these blocks as A_F. We call the blocks that are either directly below or to the right of the Toeplitz blocks "blocks adjacent to the diagonal", together they are denoted A_E. These blocks are not in general small-norm, but due to the weaker interaction, are well-suited for low-rank approximation, (see [13] for more details). Using this notation, we write our stiffness matrix as the sum of the Toeplitz blocks, blocks adjacent to the diagonal, and blocks further removed from the diagonal, $A = A_D + A_E + A_F$ (see Figure (2.1))).

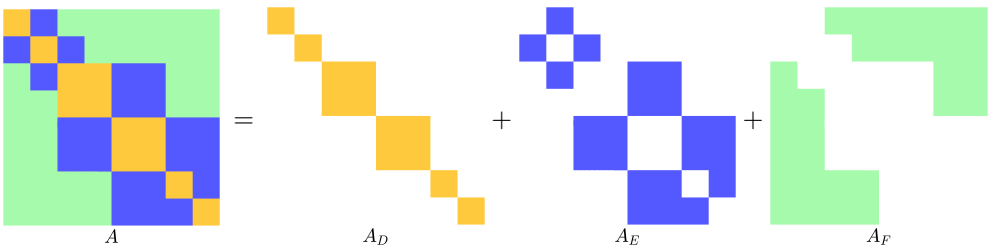

Fig. 2.1: Illustration of the stiffness matrix splitting. A_d consists of Toeplitz blocks on the diagonal, A_E i the block to the right of and below the each Toeplitz block, and A_F is everything else.

3. Adaptive $\boldsymbol{\tau}$ Preconditioner. In this section we set forth the properties we require from a preconditioner, how we use $\boldsymbol{\tau}$ in building our adaptive $\boldsymbol{\tau}$ preconditioner, which we call $\mathcal{T}$, and how to build and apply $\mathcal{T}$ efficiently.

A good preconditioner for an iterative method must in general decrease the total number of iterations, without increasing the cost of a single iteration. We borrow Kailath's [8] criteria for preconditioners, similar criteria is well-established in literature (for example [1]).

1. Complexity of constructing and applying $\mathcal{T} \in \mathbb{R}^{m\times m}$ should be $\mathcal{O}(m \log m)$.
2. A linear system with $\mathcal{T}$ should be solved in $\mathcal{O}(m \log m)$ operations.
3. The spectrum of $\mathcal{T}^{-1}A$ should be clustered around 1.

We make this last point more precise:

Definition 3.1 (Eigenvalue Clustering). *For any $\varepsilon > 0$ we say the eigenvalues of a matrix $C^{-1}L_m$ are clustered around a real number ρ if there exists N_1 and N_2 such that for all $m > N_1$ there are at most N_2 eigenvalues of $C^{-1}L_m$ that do not lie within $[\rho - \varepsilon, \rho + \varepsilon]$.*

How tightly the eigenvalues cluster around 1 determines the speed of convergence. In proving results about the clustering we use the infinite matrix framework established previously. In Section (4.1), we summarize results established previously about the spectral clustering of $\boldsymbol{\tau}$ applied to a single Toeplitz block L. This result is then used as a lemma in proving the spectral clustering for our stiffness matrix from the adaptive mesh.

To build $\mathcal{T}$, assume the stiffness matrix has k Toeplitz blocks $L_1, L_2, \ldots, L_k$ of respective sizes $m_1, m_2, \ldots, m_k$. Assume additionally these are ordered as they appear along the main diagonal. Each block has a corresponding singly infinite matrix with

a corresponding generating function, $\ell_1(z), \ell_2(z), \dots \ell_k(z)$. Assume that for all $1 \leq i \leq k, \ell_i(z) \sum_{j=-\infty}^{\infty} |t_j| < \infty$ and $\ell_i(z) > 0$ for z on the unit circle.

To explicitly construct $\mathcal{T}$ for the adaptive system, we calculate the Hankel correction H_i for every Toeplitz block, L_i. Then we define $\boldsymbol{\tau}_i = L_i - H_i$, resulting in k matrices $\boldsymbol{\tau}_1, \boldsymbol{\tau}_2, \dots, \boldsymbol{\tau}_k$. Finally, we assemble $\mathcal{T}$ as the block diagonal matrix with $\boldsymbol{\tau}_i$ as the i-th block.

$$\mathcal{T} = \begin{pmatrix} \boldsymbol{\tau}_1 & & & \\ & \boldsymbol{\tau}_2 & & \\ & & \ddots & \\ & & & \boldsymbol{\tau}_k \end{pmatrix}.$$

This explicit construction is not the most efficient, instead we apply $\mathcal{T}^{-1}$ implicitly in $\mathcal{O}\left(\sum_{j=1}^{k} m_j \log(m_j)\right) = \mathcal{O}\left(km \log(m)\right)$, where $m := \max_j m_j$. Let S_{m_i} denote the type-I DST matrix of size m_i. Using Equation (2.3) to calculate the eigenvalues of $\boldsymbol{\tau}_i$, we write $\boldsymbol{\tau}_i = S_{m_i} \Lambda_i S_{m_i}$.

(3.1)

$$\mathcal{T}^{-1} = \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix} \begin{pmatrix} \Lambda_1^{-1} & & & \\ & \Lambda_2^{-1} & & \\ & & \ddots & \\ & & & \Lambda_m^{-1} \end{pmatrix} \begin{pmatrix} S_{m_1} & & & \\ & S_{m_2} & & \\ & & \ddots & \\ & & & S_{m_k} \end{pmatrix}.$$

The DST blocks in the outside matrices can be applied via fast Fourier transform (FFT) to do matrix multiplication in $\mathcal{O}(m \log m)$ FLOPS. The eigenvalue matrix is diagonal and inverted in m operations, and can be multiplied against a Toeplitz matrix embedded in a $2m \times 2m$ circulant matrix in $\mathcal{O}(m \log m)$ FLOPS. This establishes Criteria (1) and (2) for $\mathcal{T}$.

4. Theoretical Results. In this section we prove that the spectral clustering in criteria (3) holds for $\mathcal{T}$. Proofs for clustering of circulant preconditioned problems in this area tend to follow a similar structure. First it is shown that the preconditioned system having a spectrum clustered around 1 is equivalent to the clustering of a related system around 0. This related system is then split into the sum of a low-rank term and a small-norm term. The small-norm term has spectrum clustered around 0, forcing the clustering of the entire system around 0 with the number of outliers bounded by the rank of the low-rank term.

We present our version of the spectral clustering proof for the case of $\boldsymbol{\tau}$ acting on a single Toeplitz block. This result is then used to help prove the clustering of the adaptive preconditioner on the adaptive grid, $\mathcal{T}^{-1}A$.

For convenience we state Weyl's inequality and one other lemma that is useful to us later. Both these results are corollaries to the min-max theorem.

LEMMA 4.1 (Weyl's Inequality).
Let M and E be Hermitian $m \times m$ matrices. Then for $A := M + E$ we have

$$|\lambda_k(A) - \lambda_k(M)| \leq ||E||_2 \ , 1 \leq k \leq m.$$

That is the eigenvalues of A are at most $||E||_2$ away from the eigenvalues of M.

LEMMA 4.2. *Let $T \in \mathbb{R}^{n \times n}$ be an invertible matrix. Let $A \in \mathbb{R}^{n \times n}$, then*

$$\lambda_k(T^{-1}A) \leq \frac{\lambda_k(A)}{\lambda_{\min}(T)}.$$

Proof. Using the min-max theorem,

$$\begin{aligned}\lambda_k(T^{-1}A) &= \min_{\dim V=k}\max_{x\in V}\left(\frac{(Ax,x)}{(Tx,x)}\right)\\ &\le \min_{\dim V=k}\left[\max_{x\in V}\left(\frac{(Ax,x)}{(x,x)}\right)\max_{x\in V}\left(\frac{(x,x)}{(Tx,x)}\right)\right]\\ &\le \left[\min_{\dim V=k}\max_{x\in V}\left(\frac{(Ax,x)}{(x,x)}\right)\right]\max_{x\in\mathbb{R}^n}\left(\frac{(x,x)}{(Tx,x)}\right)\\ &= \lambda_k(A)\max_{x\in\mathbb{R}^n}\left(\frac{(x,x)}{(Tx,x)}\right) \le \lambda_k(A)\frac{1}{\lambda_{\min}(T)}.\end{aligned}$$

∎

4.1. Spectral Clustering of a Toeplitz Block.

LEMMA 4.3 (Eigenvalue Clustering on a Toeplitz Block, [1, 8]). *Let L be a singly infinite, symmetric, Toeplitz matrix with diagonals $\ell_0, \ell_1, \ldots$. Then L_m is its $m \times m$ truncation with diagonals $\ell_0, \ell_1, \ldots, \ell_{m-1}$. Assume the generating function $\tilde{\ell}(z)$ is in the Wiener class and positive on the unit circle. Let $\boldsymbol{\tau}$ be the corresponding preconditioner, then for m large enough the spectrum of $\boldsymbol{\tau}^{-1}L$ is clustered around 1.*

Proof. Since $\ell(z) > 0$ for $|z| = 1$, compactness of the unit circle implies that for some $\varepsilon > 0$, $\tilde{\ell}(z) > 2\varepsilon$ when $|z| = 1$. By the Wiener class assumption we choose N such that $\sum_{j=N}^{\infty} \tilde{\ell}_j z^j \le \sum_{j=N}^{\infty} |\ell_j| < \varepsilon$. So for all $m \ge N$,

$$2\varepsilon < \ell(z) = \tilde{\ell}_m(z) + \sum_{j=m}^{\infty} \ell_j z^j < \tilde{\ell}_m(z) + \varepsilon.$$

Thus $\tilde{\ell}_m(z) > \varepsilon$ on the unit circle.

Recall that by construction $\boldsymbol{\tau} = L - H$, where H is the Hankel matrix with anti-diagonals $\ell_2, \ell_3, \ldots, \ell_{m-1}, 0, 0$. So $L = \boldsymbol{\tau} + H$ and $\boldsymbol{\tau}^{-1}L = I + \boldsymbol{\tau}^{-1}H$. Therefore it suffices to show that the eigenvalues of $\boldsymbol{\tau}^{-1}H$ are clustered around 0.

We split H into a low-rank matrix H_{LR} that contains the anti-diagonals $\ell_0, \ldots, \ell_N$ and a small-norm matrix H_{SN} such that $H = H_{LR} + H_{SN}$. Let $s := \operatorname{rank}(H_{LR}) \ll m$. The small-norm descriptor is justified since $H_{SN} = H - H_{LR}$ is a Hermitian $m \times m$ matrix with at most two copies of $\ell_N, \ldots, \ell_{m-1}$ in each row/column. Thus $||H_{SN}||_2 = \sqrt{||H_{SN}||_1 ||H_{SN}||_\infty} = ||H_{SN}||_1 < 2\varepsilon$. Hence by Weyl's Inequality at least $m - s$ of the eigenvalues of H_{SN} are clustered within 2ε of zero.

Now by Lemma (4.2),

$$\lambda_k(\boldsymbol{\tau}^{-1}H) \le \lambda_k(H)\frac{1}{\lambda_{\min}(\boldsymbol{\tau})} = \lambda_k(H)\frac{1}{\tilde{\ell}_m(z_{\min})} \le \lambda_k(H)\frac{1}{\varepsilon}. \tag{4.1}$$

Thus, the clustering of the spectrum of H around 0 implies the same for $\boldsymbol{\tau}^{-1}H$. ∎

Equation (4.1) shows that the bigger $\lambda_{\min}(\boldsymbol{\tau})$, the stronger the bound. We can get this eigenvalue exactly using Lemma (2.2), Equation (2.3). It can also be bounded in terms independent of $\boldsymbol{\tau}$, using the generating function $\tilde{\ell}(z)$, as above.

4.2. Spectral Clustering of the Full System.

THEOREM 4.4 (Full Matrix Clustering). *Let A be the stiffness matrix as described in Section (2.2) and $\mathcal{T}$ as described in Section (3). Then the spectrum of $\mathcal{T}^{-1}A$ is clustered around 1.*

Proof. Recall that the stiffness matrix is dense with Toeplitz blocks on the diagonal (denoted A_D), low-rank blocks adjacent to the diagonal (denoted A_E, and small-norm blocks elsewhere (denoted A_F), so that $A = A_D + A_E + A_F$. Let H be the block diagonal Hankel Correction matrix, observing then that $A_D = \mathcal{T} + H$ we see that $A = \mathcal{T} + H + A_E + A_F$. Our approach is to split $H + A_E + A_F$ into a small-norm term A_{SN} and a low-rank term A_{LR}. Thus

$$\mathcal{T}^{-1}A = \mathcal{T}^{-1}(\mathcal{T} + A_{SN} + A_{LR}) = I + \mathcal{T}^{-1}A_{SN} + \mathcal{T}^{-1}A_{LR}. \tag{4.2}$$

Hence, it suffices to show that the spectrum of $\mathcal{T}^{-1}A_{SN}$ is clustered around 0 and that $\mathcal{T}^{-1}A_{LR}$ contributes a bounded number of eigenvalue outliers.

Recall that blocks in A_E are well-approximated as a low-rank matrix. To that end, we begin by examining a single block in A_E, call it E (see Figure (2.1) for reference). Let E be a $m_E \times m_E$ matrix with approximation rank r_E. Then we split E via its singular value decomposition (SVD):

$$E = \left(\sum_{i=1}^{r_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top\right) + \left(\sum_{i=r_E+1}^{m_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top\right).$$

In the above equation, $\mathbf{u}$ and $\mathbf{v}$ are length m_E vectors. With a slight abuse of notation, we embed these in length m vectors to express the block matrix A_E as the SVD of its blocks. Let off-diag represent the set of off-diagonal blocks in A_E. Define $r := \max_{E \in \text{off-diag}} r_E$ and $M := \max_{E \in \text{off-diag}} m_E$. We now express A_E as the sum of the SVD of its blocks, and then regroup terms into a low-rank sum and a small-norm sum.

$$\begin{aligned} A_E &= \sum_{E \in \text{off-diag}} \left[\left(\sum_{i=1}^{r} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top\right) + \left(\sum_{i=r+1}^{m_E} \sigma_i^{(E)} \mathbf{u}_i^{(E)} (\mathbf{v}_i^{(E)})^\top\right)\right] \\ &= \left(\sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right) + \left(\sum_{i=r+1}^{M} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right). \end{aligned}$$

As before, we have $\varepsilon > 0$ and choose N much smaller than m so that for every Toeplitz block the truncated function for the block $a_{m_i}(z) > \varepsilon$ on the unit circle and $\sum_{i=N}^{\infty} |a_i| < \varepsilon$. We additionally split H by separating into H_{LR} the anti-diagonals with coefficients $a_0, \ldots, a_N$ and into H_{SN} the anti-diagonals comprising of $a_{N+1}, \ldots a_{m_i}$ for each block. So we have two splittings,

$$\begin{aligned} A_E &= \left(\sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right) + \left(\sum_{i=r+1}^{M} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top\right), \\ H &= H_{SN} + H_{LR}. \end{aligned}$$

The first term in each sum can be thought of as the low-rank equivalent from before and similarly the second term is the small-norm summand. Finally, we make the splitting $A = \mathcal{T} + A_{SN} + A_{LR}$, where

$$\begin{aligned} A_{SN} &= H_{SN} + \sum_{i=r+1}^{M} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top + A_F, \\ A_{LR} &= H_{LR} + \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top. \end{aligned}$$

Now $s := \text{rank}(\mathcal{T}^{-1}A_{LR}) \leq \text{rank}(A_{LR}) \leq 2(N-1)+r$ bounds the number of outlier eigenvalues. It remains to show $||\mathcal{T}^{-1}A_{SN}||_2 \leq \varepsilon$. Define $\tilde{A_E} = \sum_{i=r_{A_E}+1}^{M} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$, so that $A_{SN} = H_{SN} + \tilde{A_E} + A_F$, then

$$||\mathcal{T}^{-1}A_{SN}||_2 \leq ||\mathcal{T}^{-1}H_{SN}||_2 + ||\mathcal{T}^{-1}\tilde{A_E}||_2 + ||\mathcal{T}^{-1}A_F||_2.$$

Since the block multiplication $\mathcal{T}^{-1}H_{SN}$ is each $\boldsymbol{\tau}$ acting on its corresponding Hankel correction, it is bounded by Lemma (4.3). We bound $||\mathcal{T}^{-1}A_F||_2$ with Weyl's inequality:

$$||\mathcal{T}^{-1}A_F||_2 \leq ||\mathcal{T}^{-1}||_2||A_F||_2 = \sigma_{\max}(\mathcal{T}^{-1})\sigma_{\max}(A_F) = \frac{\sigma_{\max}(A_F)}{\lambda_{\min}(\mathcal{T})}.$$

Since A_F are small-norm this value is small. Finally, we bound $||\mathcal{T}^{-1}\tilde{A_E}||_2$ using Lemma (4.2):

(4.3)
$$\lambda_k(\mathcal{T}^{-1}\tilde{A_E}) \leq \lambda_k(\tilde{A_E})\frac{1}{\lambda_{\min}(\mathcal{T})} = \lambda_k(\mathcal{T}^{-1}\tilde{A_E}) \min_i \left(\tilde{\ell}_i(z_{\min})\right) \leq \lambda_k(\mathcal{T}^{-1}\tilde{A_E})\frac{1}{\varepsilon}.$$

To finish the proof, we utilize the fact that the matrices are SPD and relate the k-th eigenvalue of the preconditioned matrices to their 2-norm,

$$\begin{aligned} ||\mathcal{T}^{-1}A_{SN}||_2 &\leq ||\mathcal{T}^{-1}H_{SN}||_2 + ||\mathcal{T}^{-1}A||_2 + ||\mathcal{T}^{-1}A_F||_2 \\ &= \lambda_{\max}(\mathcal{T}^{-1}H_{SN}) + \lambda_{\max}(\mathcal{T}^{-1}\tilde{A_E}) + \lambda_{\max}(\mathcal{T}^{-1}A_F) \\ &\leq \frac{1}{\lambda_{\min}(\mathcal{T})}\left(\lambda_{\max}(H_{SN}) + \lambda_{\max}(\tilde{A_E}) + \lambda_{\max}(A_F)\right). \end{aligned}$$ □

The strength of this bound depends on $\lambda_{\min}(\mathcal{T})$, but can be written independently of $\mathcal{T}$ in terms of the generating functions as in Equation (4.3). This dependence is the same for the single block case, and appears in similar proofs for circulant preconditioners (see for example [8]).

In practice the block size, m_i, is determined by the AFEM algorithm, which in turn dictates the actual size ε and hence the strength of clustering. So bigger Toeplitz blocks in A result in tighter eigenvalue clustering.

Notice that extending the proof from one Toeplitz block to the adaptive stiffness matrix relies only on being able to split the block into a small-norm and a low-rank piece. Thus, nothing about this proof is overly-specific to $\boldsymbol{\tau}$ and the methods could easily be extended to the other types of circulant preconditioners in [3, 4, 5].

5. Numerical Results.

5.1. Validating Assumptions. In order to prove the spectral clustering, we made two assumptions about the structure of A: the blocks adjacent to the diagonal are well-approximated by low-rank matrices and the other off-diagonal blocks are small-norm. Table (5.1) displays the maximum and average (median) rank needed to achieve an approximation under ε on each block. Table (5.2) shows the norm and relative norm of $||A_F||_2$. Both tables include results for $\alpha = 1.3, 1.5, 1.9$, after 7 levels of refinement, but similar trends were seen across all fractional orders and refinement levels. In the case of 1×1 Toeplitz blocks we found that including the entry to the right and underneath in A_E was not sufficient to capture all the low-rank information. Instead, the corresponding block in A_E was extended to dimensions 5×5 such that the original entry is the center of that block. In terms of spectral clustering, this could increase the number of outliers, but better characterizes the size of the cluster.

ε \ α	1.3	1.5	1.9
10^{-3}	4(5)	4(6)	5(6)
10^{-4}	5(7)	5(7)	6(8)
10^{-5}	6(8)	6(9)	7(9)

Table 5.1: This table shows the approximation rank needed for the error to be less than ε for varying fractional indices α. Table entries are displayed as `median (max)`.

α	1.3	1.5	1.9
$\|\|A_F\|\|_2$	0.6138	1.528	4.479
$\frac{\|\|A_F\|\|_2}{\|\|A\|\|_2}$	4.893×10^{-2}	2.368×10^{-2}	2.533×10^{-3}

Table 5.2: This table shows the norm of $||A_F||_2$ and $||A_F||_2/||A||_2$ across values of α.

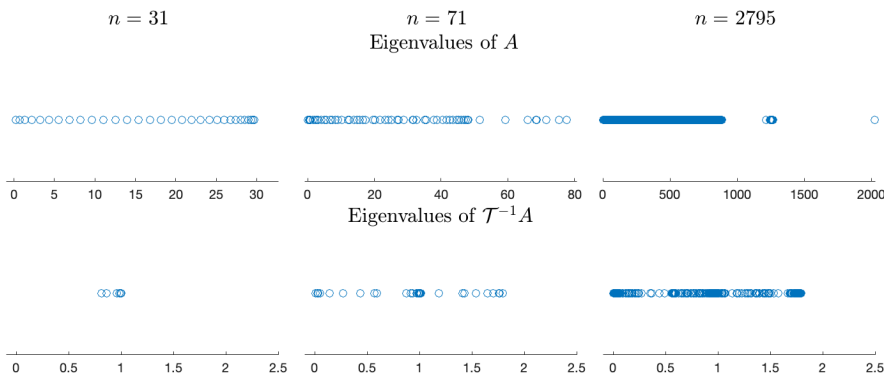

Fig. 5.1: Eigenvalue clustering before and after preconditioning. Each column represent a different matrix size. The first row shows the spectrum before preconditioning, the second shows the spectrum at the same scale after preconditioning. The third row shows the preconditioned spectrum on a smaller interval.

5.2. Improved Convergence. In Figure 5.1, we see some example results of the preconditioner on the spectrum of A. These tests are for the problem (1.1) with $\alpha = 1.7$ and exact solution $u(x) = 10x^2(1-x^2)$ on $[0, 1]$ (see [13] example 6.1 for more details). The three different matrix sizes are $n = 31, 71, 2795$ and reflect 0, 3, and 11 levels of refinement respectively. Note that zero levels of refinement corresponds to a uniform mesh.

The left-most column represents a matrix that consists of a single Toeplitz block, so the preconditioner reduces to the case of $\boldsymbol{\tau}$. In this case the clustering of the spectrum around 1 is very strong. In the $n = 71$ case the biggest Toeplitz blocks are size 11, and there are two such blocks. The eigenvalue clustering around 1 is still observable, but the size of the cluster has grown. Finally for $n = 2795$, the preconditioner reduces the spectral radius by several orders of magnitude. The spectrum is concentrated around 1, but the distribution of eigenvalues between 0 and 2 is more uniform. For this matrix the biggest Toeplitz block is size 939 and there are two other

blocks just bigger than 200. On the other hand, there are 245 1×1 blocks. For all cases, the strength of the clustering is explained by the amount of smaller Toeplitz blocks in A. Notice that although the proof allows for eigenvalue outliers beyond the cluster, in practice everything remains within the interval $(0, 2)$.

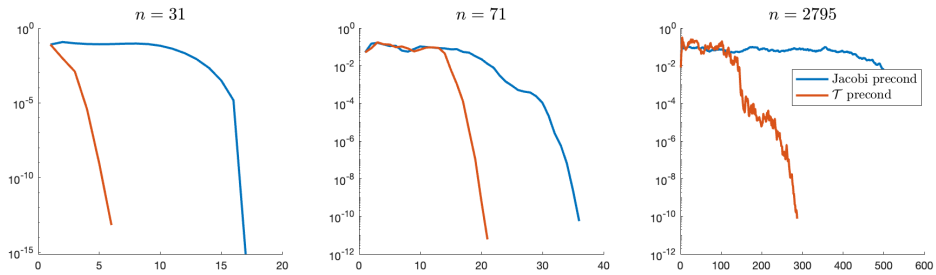

Fig. 5.2: Comparing the residual vector norm across PCG iterations. The blue line uses a Jacobi preconditioner and the orange line uses our $\mathcal{T}$ preconditioner.

We see how the clustered spectrum translates to accelerated convergence. In Figure (5.2) the same matrices are used for preconditioned conjugate gradient (PCG), comparing the adaptive $\boldsymbol{\tau}$ preconditioner and a Jacobi preconditioner with a tolerance of 10^{-8}. For the two smaller matrices the preconditioner clearly accelerates the point of superlinear convergence [11]. For the final matrix the Jacobi-preconditioned system fails to converge within the maximum of 600 iterations, but with the $\mathcal{T}$ preconditioner converges in about 300 iterations.

6. Conclusion. We have shown how to build a preconditioner for FDEs discretized by AFEM. The spectrum of the preconditioned system is clustered around 1 and accelerates convergence of iterative methods. In the future, we will investigate how to customize the AFEM `MARK` module to build bigger Toeplitz blocks and thus get tighter clusters. Finally, we will investigate how to use these techniques to build preconditioners for FDEs discretized with AFEM on higher dimensional meshes.

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